
Presentation Playbook — how to actually deliver this

Everything about *giving* the talk: timing, who-says-what, rehearsal, the demo, handling questions, examiner psychology, and what gets you marks. Pair with `SPEAKER_NOTES.md` (per-slide script) and `DISCUSSION_QA.md` (answers).

1. The format, decoded

- **15–20 min talk + 5–10 min discussion.** Plan the talk for **18 min** — it leaves slack and you won't get cut off. Over-running looks unprepared.
- **20 slides** → ~50 s/slide average. Some (title, team, demo) are 15–20 s; the technical ones (8–12) are 60–90 s. That balances out to ~18 min.
- **Three speakers** — examiners expect everyone to talk roughly equally. The split below gives each person ~6 min + their own Q&A territory.

2. Speaker split (rehearse these boundaries)

Speaker	Owens	Slides	~min
Emir (infra/data)	task, story, data, engineering, status, takeaways	2, 3, 7, 14, 15, 19	~6
Dima (model)	VITS, components, MAS, the challenge, forward_train, losses, discriminator, GAN loop	4, 5, 6, 8, 9, 10, 11, 12	~8
Ilya (eval)	evaluation, failure modes, (co-drive demo)	13, 16, 17	~4

Title (1), team (18), thank-you (20) — whoever. **Hand-offs out loud:** “...and Dima will take the model.” A clean verbal baton looks rehearsed.

Dima carries the technical core — that’s by design, it’s the meat. If you want a more even split, give Dima’s slides 5 (components) and 6 (MAS) to whoever is most comfortable with the VAE/alignment story.

3. The narrative arc (so it’s a story, not a file dump)

1. **Hook** — “TTS usually needs two models; we used one, end-to-end: VITS.” (2–4)
2. **Twist** — “and we built our own voice dataset so you can *hear* it work.” (7)
3. **The hard part** — “Hugging Face only ships VITS for inference, so we wrote the entire training loop ourselves.” (8–12) ← *this is your differentiator*
4. **Proof** — “here’s how we measure it, and here’s the baseline.” (13)
5. **Honesty** — “everything’s built; the GPU run is the last step.” (15)
6. **Payoff** — play the audio. (17)

Every slide should advance this arc. If a slide doesn’t, cut it.

4. Rehearsal checklist (do this twice, out loud, timed)



Full run-through with a timer; note the minute mark at slide 13 — should be ~16:00. If you hit it at 18:00, you’re too slow; trim slides 9–11.



Each speaker can give **their** slides without reading them.



Hand-offs are verbal and smooth.



Demo dry-run on the presentation machine (see §6) — audio actually plays through the room speakers.



One person drives slides the whole time (the others don’t fight for the clicker).



Decide who fields which Q&A area (model Qs → Dima, eval Qs → Ilya, data/eng → Emir). Say so beforehand so you don’t talk over each other.



Have `slides.pdf` AND `slides.pptx` AND `slides.html` on a USB stick / in cloud — projectors are hostile.

5. Per-slide “what they’ll fixate on”

Quick map of where questions cluster, so you pre-load the answer:

Slide	Likely question	One-line defense
3 (pivot)	“Why abandon SpeechT5?”	One model > two; better audio out of the box.
7 (data)	“Synthetic data — isn’t that circular?”	Demonstrates <i>adaptation</i> ; real recordings would slot into the same pipeline.
8 (challenge)	“What exactly did you implement?”	forward_train, MAS, 5 losses, MPD+MSD, GAN loop.
9 (forward_train)	“Why segment-slice?”	Decoder is expensive; HiFi-GAN trick; convolutional so a crop is fine.
10 (losses)	“Why 45× on mel?”	Spectral accuracy is primary; GAN terms are polish.
13 (eval)	“Why Whisper isn’t circular?”	Independent ASR; its error floor is constant across before/after.
15 (status)	“So it doesn’t work yet?”	Machinery is done + wired; only the multi-hour run remains.

Full answers: `DISCUSSION_QA.md` .

6. The demo — don’t let it kill you

Live demos fail in front of examiners. **Pre-render everything:**

- Have the wavs ready in a folder and a simple player open: `dl/audio/generated/test_0000.wav` (pretrained), `dl/droid_test.wav` (B1 voice), `dl/audio/demo_4_excited.wav` (a fun effect).
- Order: **(1) pretrained VITS** → **(2) B1 droid voice** → **(3) one effect**. Three clips, ~10 s each. That’s the whole point made audible.
- **Backup:** if audio fails, the slide text still tells the story — say “you’ll hear the droid voice in the recording we’ll share.” Never spend >20 s fighting hardware.
- **Test the room’s audio output before you start.** Mac volume up, correct output device, file associations work.

7. Handling questions (the 5–10 min discussion)

- **Repeat/rephrase the question** before answering — buys thinking time, makes sure you understood, and the room hears it.
- **Answer in one sentence, then offer depth.** “Yes — because X. Want the detail?” Examiners reward crisp answers over rambling.
- **If you don’t know:** “We didn’t measure that, but I’d expect ..., because” Reasoning from principles beats bluffing. Never invent a number.
- **Route by ownership** — the model person takes model questions, etc. But don’t leave a teammate hanging; jump in if they stall.
- **The mock results trap.** If asked about `training_log.json`, say plainly: “that’s a placeholder from the old SpeechT5 plan — not real results. Our real baseline is the pretrained VITS row: WER 5.5%.”
- **Turn it around (good):** “We considered phonemes here — is that the direction you’d push?” Shows you know the trade-space.

8. What earns marks (rubric alignment)

The brief asks for specific things — make sure each is *visibly* covered:

Brief requirement	Where you show it	Say the words
Chosen pipeline justified	slides 2–4	“Option C, VITS, end-to-end.”
Trained on held-out test set	slides 12–13	“held-out split / test manifest.”
Training budget (epochs/ batch/GPU-h)	slide 12	“batch 16, lr 2e-4, N epochs, ~X GPU-hours.”
Qualitative gallery	slide 17 (demo)	play the clips.
Failure modes	slide 16	“mispronunciation, prosody, artifacts.”
ASR → WER/CER	slide 13	“Whisper, WER 5.5% / CER 1.6%.”
Spectral metric (MCD)	slide 13	“MCD 6.38 dB.”
Optional: tokenizer study	slide 16 / Q&A	“char now; phonemes the obvious extension.”

If you tick every row out loud, you’ve covered the assignment explicitly — that’s the difference between a good and a top mark.

9. Slide-design sanity (already done, keep it)

- One idea per slide, ≤ 6 bullets, big font (the deck is 26 px body / 46 px H1).

- Tables and the pipeline diagram do the heavy lifting — point at them, don't read them.
- Don't add a wall of equations to the slides — keep the math in `TECHNICAL_APPENDIX.md` and pull it out *only* if asked.

10. Backup / “deep-dive” slides to have ready (optional)

If you expect a hard technical panel, keep 2–3 hidden slides after the thank-you (or just know them from the appendix):

- **MAS in one diagram** — the DP grid + monotonic path.
- **The ELBO** — $\log p(x) \geq \mathbb{E}[\log p(x|z)] - \text{KL}(q||p)$ mapped to your losses.
- **MPD reshape** — 1-D audio $\rightarrow$ 2-D by period, why primes.

These never appear unless asked; they signal depth when they do.

11. Logistics morning-of



Laptop charged + charger; HDMI/USB-C adapter.



`slides.pdf` open and tested on the projector resolution (16:9).



Audio tested through room speakers.



Phones on silent; one person owns the clicker.



Water. Breathe. You built a from-scratch VITS trainer — own it.

12. The 30-second version (if you're cut to almost nothing)

“We built end-to-end TTS by fine-tuning VITS onto a custom voice we created and published. Hugging Face only ships VITS for inference, so we re-implemented the whole training stack ourselves — alignment search, five losses, the HiFi-GAN discriminators, the GAN loop. We evaluate with Whisper-based WER/CER for intelligibility and MCD for spectral fidelity. The pipeline is built end-to-end; the final fine-tuning run is the last step.”